GROUP 9
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PERSONALIZED EXPOSURE ESTIMATION

POLITECNICO MILANO 1863

RESEARCH QUESTIONS

Q1: Are the stations statistically different in terms of pollutants levels?

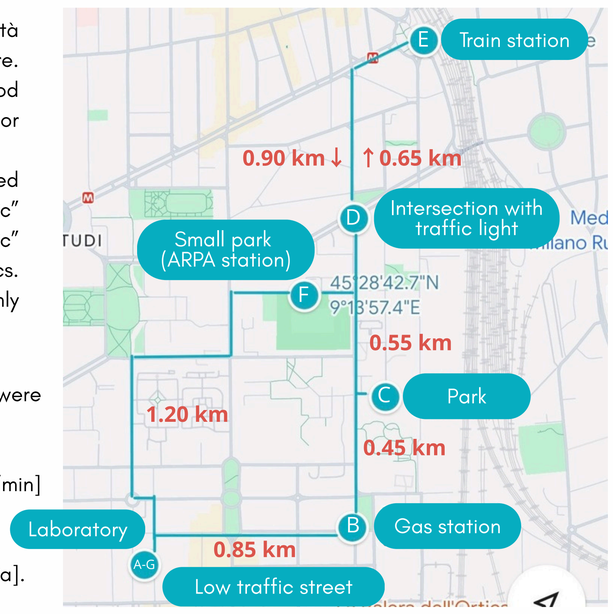
Q2: How does pollution influence Heart Rate?

Q3: Are there other significant drivers?

DATASET DESCRIPTION

20 healthy subjects followed a walking route inside the Città Studi district at the end of November, as shown in the figure. Participants started at Point A (the laboratory), where they stood still for 5 minutes, and then walked at a normal pace, stopping for 5 minutes at each of the predefined locations.
For each participant environmental parameters were recorded continuously, with a wearable device, during walking ("dynamic" phase) and while standing still at the selected locations ("static" phase) each with distinct environmental characteristics. Physiological parameters on the other hand were registered only in the static phases once a minute for 5 minutes.
The recorded parameters include:

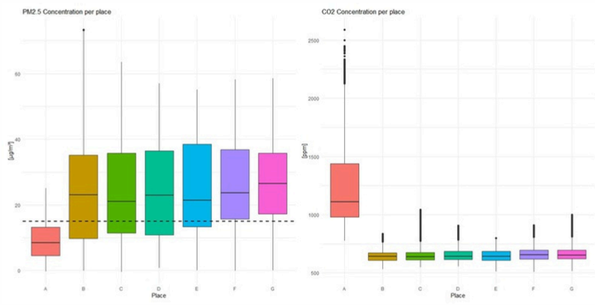
- Particulate matter: **PM1**, **PM2.5**, **PM10** (PM1 and PM10 were derived from PM2.5 measurements) [$\mu g/m^3$];
- Gas (**CO_2**) [ppm];
- Physiological signals: breathing frequency (**FB**) [breaths/min] and heart rate (**HR**) [bpm];
- Volatile Organic Compounds (**VOC**) [ppb];
- Temperature (**T**) [°C], Humidity Rate (**RH**) [%], Pressure (**P**) [Pa].



PRELIMINAR ANALYSIS

Point A has significantly different values of pollutants with respect to all the other outdoor places; this is probably due to lack of ventilation and human presence that increase levels of **CO_2**, while the absence of traffic explains why **PM2.5** is lower.
All locations' concetrations of **PM2.5** consistently exceed the legal limit of 15 $\mu g/m^3$.
A few extreme values, both high and slightly negative were found. The latter are likely due to sensor drift or noise, especially when actual concentrations are close to zero.
VOC values were generally low but showed unexpected peaks at location G (a low-traffic square). We will not include it in our analysis since it is a mixture of too different types of chemical compounds.
Temperature readings are consistently higher indoors (lab) than outdoors, which is coherent with the experiment being conducted in late November.
They are also systematically higher than expected:

- 27°C indoors vs expected 22°C ;
- 21°C outdoors vs expected 10°C.



	min	1st Q.	median	mean	3rd Q.	max
CO2	509.1	618.4	656.6	737.1	711.2	2588.1
PM 2.5	-0.375	10.3	20.0	22.3	34.1	73.3
VOC	-831.3	89.9	200.8	315.8	461.9	11361.5
T	10.0	18.5	22.0	22.2	26.0	35.1
RH	31.9	44.0	50.0	49.4	55.0	70.7
P	91950	100112	100824	100766	101439	123420

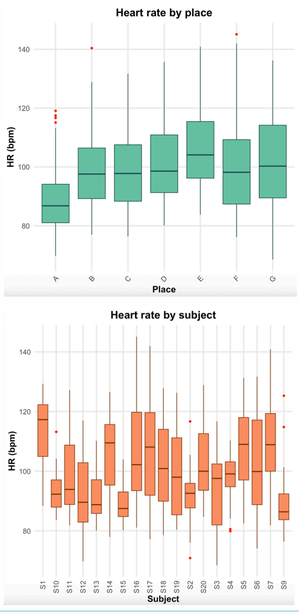
ANOVA ANALYSIS ON HR

Note: for all our analyses from now on, we excluded Place A and Subject 8 from the dataset for the following reasons:

- Place A is an indoor environment and so consistently exhibited lower heart rate (HR) values across all subjects. Including it could have introduced a structural bias;
- Subject 8 showed irregular and inconsistent recordings, likely resulting from sensor malfunction or improper device calibration. Retaining this subject would have compromised the reliability of the results.

1. One-Way ANOVA (Response: HR, Factor: Place):
After removing Place A, the factor place is not statistically significant.
2. One-Way ANOVA (Response: HR, Factor: Subject):
The analysis confirms a significant effect of subject on HR.
This is consistent with physiological differences in baseline heart rate between individuals.

Assumptions: For all models, assumptions of normality and homogeneity of variances were verified and reasonably satisfied. Box-Cox trasnformations were applied.

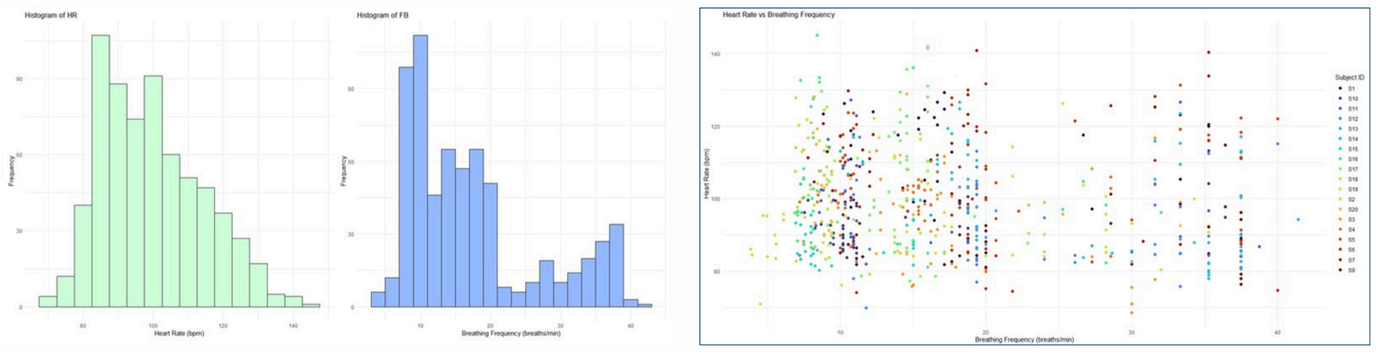


ANOVA ANALYSIS ON PM2.5 AND CO2

We performed an ANOVA analysis to see whether outdoor places have statistically significant concentrations of **PM2.5** and **CO2**. Several strategies were adopted to recover gaussianity (Box-Cox, Scaling and Yeo-Johnson) but they did not produce the expected result. Despite this, the answer is negative (P-value $\simeq$ 1), which is coherent with previous boxplots.

DATA PROCESSING

Since **HR** and **FB** were measured once a minute we only have 5 observations for each subject in each place. For this reason it was necessary to aggregate enviromental data to go from 300 observetions per subject per place to 5 observations.
We assumed that the first HR measurement corresponds to time zero, that is, the exact moment when each subject arrives at a station and begins the static phase. Therefore, we associated the first observations of each station with the average pollutant values recorded during the final moments of the dynamic phase. The remaining four observations are associated with the average pollutant values recorded during the preceding minute.



Other factors that could influence HR and FB are the distance traveled to reach a given station and time. Thus, we created two variables:

- **cum.dist.:** tracks the number of kilometers covered to reach the station where the subject is located, starting from Place A. The distances between the various stations were determined by reconstructing the route using Google Maps.
- **min.still**: keeps track of the number of minutes elapsed since arriving at the current station, which takes values in {0, ..., 4}.

The graph on the top-right shows that given a fixed subject and a value of breathing frequency (FB), heart rate (HR) has too high variability.
Furthermore, we noticed that the device appears to lose sensitivity above 20 breaths per minute. In fact, it seems that beyond this limit, FB mainly takes integer values.
Note: FB was indirectly measured by three Inertial Measurement Units (IMUs) by tracking small body movements that occur during breathing while HR was directly measured.

HEART RATE MODELLING

$$\log(HR_{ij}) = \beta_0 + \beta_1 \cdot min.still_{ij} + \beta_2 \cdot \log(PM2.5_{ij}) + \beta_3 \cdot \log(CO2_{ij}) + \beta_4 \cdot RH_{ij} + \beta_5 \cdot cum.dist._{ij} + \beta_6 \cdot T_{ij} + \beta_7 \cdot (min.still_{ij} \times PM2.5_{ij}) + b_{0i} + \varepsilon_{ij}$$

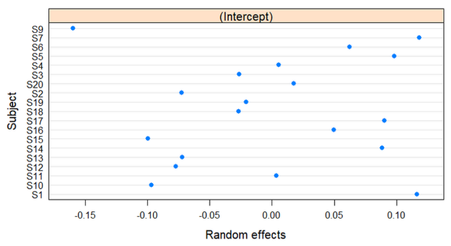
Fixed Effects

Variable	Estimate
(Intercept)	5.232^{***} (0.0000)
min.still	-0.044^{***} (0.0000)
$\log(PM_{2.5})$	-0.009 (0.6484)
$\log(CO_2)$	-0.090 (0.3083)
RH	0.001 (0.5442)
cum.dist	-0.002 (0.6570)
T	-0.001 (0.8360)
min.still × PM2.5	0.001^{**} (0.0023)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Random Effects

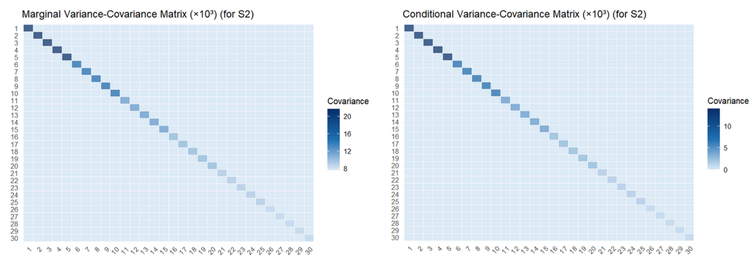
$b_{0i} \sim N(0,\sigma_b^2)$ is the individual random effect on the intercept where $\sigma_b^2 = 0.00764$



PVRE = 0.4072

Variance function

$\epsilon_{ij} \sim N(0,\sigma^2 w_{ij}^2)$ with $Var(\epsilon_{ij}) = \sigma^2 |cum.dist._{ij}|^{2\delta}$ and $\sigma^2 = 0.011127$

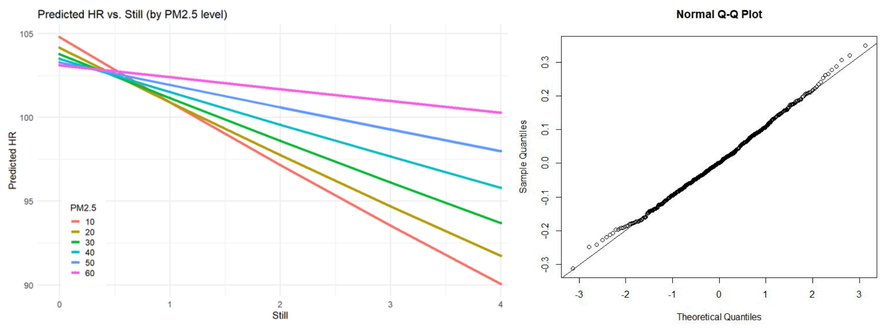


Model interpretation

The higher the pollutant level, the less sensitive HR becomes to additional increases. It is reasonable to assume that at higher concentrations the response plateaus due to physiological limits. Moreover, it was observed that HR variance increases with its mean and log transformation helps flattening that.

The main results of our model is that the interaction between min.still and PM2.5 (β_7 = 0.001) is positive, indicating that the recovery from fatigue is slowed down by the time spent in a polluted place.

As visible in the HR boxplots, individual physiological variability leads to a high value of PVRE. Normality of residuals is verified since the P-value associated to the Shapiro-Wilk test is equal to 0.3052.



STARTING MODEL

LMM model for log(HR) with random effects on subject, considering only outdoor data.

Covariates: log(PM2.5), log(CO2), cum.dist., T, RH.

Problems: nothing is significant, residuals are normally distributed and heteroskedastic.

UPDATED MODEL

LMM model for log(HR) with random effects on subject, considering only outdoor data.

Additional covariates: min.still and its interaction with PM2.5.

Problems: residuals are heteroskedastic.

FINAL MODEL

LMM model for log(HR) with random effects on subject, considering only outdoor data.

Covarietes: same as before.

Adding **VarPower** using cum.dist as covariate and subject as the grouping factor.

ANSWERS AND CONCLUSIONS

Question 1

As we can see from the ANOVA analysis on the pollutants, their levels are not significantly different.
Knowing that the experiment takes place in a small area and the relative distance between locations is not sufficient to observe relevant fluctuations, this conclusion seems reasonable.

Question 2

Pollutants do not have an impact on HR on their own but they become significant only when considering their joint effect with time.
This conclusion is consistent since we are more affected by pullutants when we breathe them for a prolonged time period.

Question 3

At first glance, one might think that walked distance, temperature and humidity rate would be significant, but this is not verified.
The second main driver is the heterogeneity within the subjects, as visible in ANOVA and random effect results.

Limitations and further developments

It would have been nice to examine in depth the different behaviour between male and female subjects, as well as the impact of their age on the HR.
In addition the lack of information on the days when the experiment was conducted limited our possibility to compare our results with external sources.
A possible improvement to the analysis would be refining the precision of the wearable device and repeating the same experiment in different seasons.

TUTORS: *Proff. Lara Cavinato and Alessandra Ragni*

REFERENCE: *S. Bernasconi, A. Angelucci, A. Rossi, A. Alimberti. A wearable system for personal air pollution exposure: a walk-about in Milan. 62 (suppl. 67), 2023*